

Arjuna JEE 2025

Practice Sheet (Mathematics)

Sequence and Series

- Q1** The sum of the first n terms of two arithmetic series are in the ratio $(7n + 1) : (4n + 27)$. The ratio of their 11^{th} terms is
 (A) 2 : 3 (B) 3 : 2
 (C) 3 : 4 (D) 4 : 3
- Q2** If $S_1, S_2, S_3, \dots, S_p$ are the sums of first n terms of p arithmetic progressions whose first terms are $1, 2, 3, \dots, p$ and whose common differences are $1, 3, 5, 7, \dots$, then the value of $S_1 + S_2 + S_3 + \dots + S_p$ is
 (A) $np(np + 1)/2$
 (B) $np(np - 1)/2$
 (C) $(np - 1)(np + 1)/2$
 (D) $n(np + 1)$
- Q3** The number of terms in an AP is even. The sum of the odd terms is 24 and the sum of the even terms is 30. If the last term exceeds the first term by $10\frac{1}{2}$, then the number of terms in the AP is
 (A) 6 (B) 8
 (C) 10 (D) 12
- Q4** The m^{th} term of an AP is a and its n^{th} term is b , and $m \neq n$. Then the sum of the first $(m + n)$ terms of the AP is
 (A) $\frac{m+n}{2} \left[a + b + \frac{a-b}{m-n} \right]$
 (B) $\frac{m+n}{2} \left[(a + b) - \frac{a-b}{m-n} \right]$
 (C) $\frac{m+n}{2} \left[(a + b) + \frac{a-b}{m-n} \right]$
 (D) $\frac{m+n}{2} \left[(a - b) - \frac{a-b}{m-n} \right]$
- Q5** If the numbers $3^{2a-1}, 14, 3^{4-2a}$ ($0 < a < 1$) are the first three terms of an AP , then its fifth term is equal to.
 (A) 33 (B) 43
 (C) 53 (D) 63
- Q6** In a sequence $a_1, a_2, a_3, \dots$ of real numbers it is observed that $a_p = \sqrt{2}, a_q = \sqrt{3}$ and $a_r = \sqrt{5}$, where p, q, r are positive integers such that $1 \leq p < q < r$. Then
 (A) a_p, a_q, a_r can be terms of an AP
 (B) $1/a_p, 1/a_q, 1/a_r$ can be terms of an AP
 (C) a_p, a_q, a_r can be terms of an AP if and only if p, q, r are perfect-squares
 (D) Neither a_p, a_q, a_r are in AP nor $1/a_p, 1/a_q, 1/a_r$ are in AP
- Q7** If the sum of the roots of the quadratic equation $ax^2 + bx + c = 0$ is equal to the sum of the squares of their reciprocals, then
 (A) bc^2, ca^2, ab^2 are in AP
 (B) bc^2, ab^2, ca^2 are in AP
 (C) ca^2, bc^2, ab^2 are in AP
 (D) ab, bc, ca are in AP
- Q8** If the reciprocals of the roots of the equation $10x^3 - ax^2 - 54x - 27 = 0$ are in AP , then the value of a is
 (A) 6 (B) 8
 (C) -9 (D) 9
- Q9** If $5^2 \cdot 5^4 \cdot 5^6 \dots 5^{2n} = (0.04)^{-28}$, then n is equal to
 (A) 7 (B) 5
 (C) 6 (D) 3
- Q10** The difference between the fourth and the first term of a GP is 52 and the sum of the first three terms is 26. Then the sum of the first six terms is
 (A) 720 (B) 725
 (C) 728 (D) 780
- Q11**



A total of n arithmetic means are inserted between x and $2y$ and further n arithmetic means are inserted between $2x$ and y . If the k^{th} arithmetic means of both sets are equal, then a relation between x and y is

- (A) $ky = (n - k)x$
 (B) $ky = (n + 1 - k)x$
 (C) $k(y + 1) = (n - k)x$
 (D) $k(y + 1) = (n - 1 - k)x$

Q12 If the sum of the first n terms of an AP is cn^2 , then the sum of the squares of these terms is

- (A) $\left[n(4n^2 - 1)c^2 \right] / 6$
 (B) $\left[n(4n^2 + 1)c^2 \right] / 3$
 (C) $\left[n(4n^2 - 1)c^2 \right] / 3$
 (D) $\left[n(4n^2 + 1)c^2 \right] / 6$

Q13 In an AP , if s_n is the sum of the first n terms (n is odd) and S'_n is the sum of the terms of the series in odd places then $s_n / s'_n =$

- (A) $2n/n + 1$
 (B) $n/n + 1$
 (C) $n + 1/2n$
 (D) $n + 1/n$

Q14 The arithmetic mean of two numbers is $18\frac{3}{4}$ and the positive square root of their product is 15. The larger of the two numbers is

- (A) 24 (B) 25
 (C) 20 (D) 30

Q15 Three numbers are in AP . If 8 is added to the first number, we get a GP with sum of the terms is equal to 26. Then the common ratio of the GP when they are written in the ascending order, is

- (A) 3 (B) $1/3$
 (C) 2 (D) $1/2$

Q16 Let x and y be real numbers such that $x, x + 2y$ and $2x + y$ are in AP and $(y + 1)^2, xy + 5$ and $(x + 1)^2$ are in GP . With this information, answer the following questions. The common difference of the AP is

- (A) 2 (B) 3

- (C) 4 (D) $3/2$

Q17 Let x and y be real numbers such that $x, x + 2y$ and $2x + y$ are in AP and $(y + 1)^2, xy + 5$ and $(x + 1)^2$ are in GP . With this information, answer the following questions.

The common ratio of the GP is

- (A) 2 (B) $1/2$
 (C) 3 (D) $1/3$

Q18 Let x and y be real numbers such that $x, x + 2y$ and $2x + y$ are in AP and $(y + 1)^2, xy + 5$ and $(x + 1)^2$ are in GP . With this information, answer the following questions. The sum of four AMs between x and y and product of four GMs between x and y is

- (A) $4 + 4\sqrt{3}$
 (B) $4(2 + \sqrt{3})$
 (C) $8(1 + \sqrt{3})$
 (D) 17

Q19 Match the items in column I with those in column II.

	Column-I	Column - II
(A)	If the $p^{\text{th}}, q^{\text{th}}$ and r^{th} terms of an AP are a, b, c respectively, then the value of $a(q - r) + b(r - p) + c(p - q)$ is	(p) 15
(B)	The sum of m terms of an AP is n and the sum of n terms is m , then the $(m + n)$ terms is	(q) 27
(C)	If five arithmetic means are inserted between 2 and 4, then the sum of the five means are	(r) $-(m + n)$
(D)	In an AP , if the sum of n terms is $3n^2$ and the sum of m terms is $3m^2$ ($m \neq n$) then, the sum of the first three terms is	(s) 0

- (A) (A) $\rightarrow$ (s), (B) $\rightarrow$ (p), (C) $\rightarrow$ (p), (D) $\rightarrow$ (r)
 (B) (A) $\rightarrow$ (q), (B) $\rightarrow$ (r), (C) $\rightarrow$ (q), (D) $\rightarrow$ (p)
 (C) (A) $\rightarrow$ (s), (B) $\rightarrow$ (r), (C) $\rightarrow$ (p), (D) $\rightarrow$ (q)
 (D) (A) $\rightarrow$ (r), (B) $\rightarrow$ (s), (C) $\leftrightarrow$ (p), ((D) $\rightarrow$ (q))

Q20 Match the items in column I with those in column II.



Column-I		Column - II	
(A)	The sum of all integers between 250 and 1000 which are divisible by 3 is	(p)	3050
(B)	The sum of all odd numbers between 1 and 1000 that are divisible by 3 is	(q)	156375
(C)	The sum of all integers from 1 to 100 which are divisible by exactly one of 2 and 5 is	(r)	3550
(D)	If $7100A$'s are inserted between $\sin^2 \theta$ and $\cos^2 \theta$, then their sum is	(s)	83667

- (A) (A) $\rightarrow$ (r), (B) $\rightarrow$ (q), (C) $\rightarrow$ (s), (D) $\rightarrow$ (p)
 (B) (A) $\rightarrow$ (s), (B) $\rightarrow$ (s), (C) $\rightarrow$ (r), (D) $\rightarrow$ (p)
 (C) (A) $\rightarrow$ (q), (B) $\rightarrow$ (s), (C) $\rightarrow$ (p), (D) $\rightarrow$ (r)
 (D) (A) $\rightarrow$ (p), (B) $\rightarrow$ (r), (C) $\rightarrow$ (s), (D) $\rightarrow$ (p)

Q21 In column I some series are given and in column II their n^{th} terms are given. Match them.

Column - I		Column - II	
(A)	$3/4 + 5/36 + 7/144 + 9/400 + \dots$	(p)	$3^{n-1} + n$
(B)	$2 + 5 + 12 + 31 + 86 + 249 + \dots$	(q)	$1/6(n^3 + 6n^2 + 11n + 6)$
(C)	$4 + 10 + 20 + 35 + 56 + 84 + 120 + \dots$	(r)	$1/2(n^2 - n + 2)$
(D)	$1 + 2 + 4 + 7 + 11 + \dots$	(s)	$(2n + 1)/n^2(n + 1)^2$

- (A) (A) $\rightarrow$ (s), (B) $\rightarrow$ (p), (C) $\rightarrow$ (r), (D) $\rightarrow$ (q)
 (B) (A) $\rightarrow$ (s), (B) $\rightarrow$ (p), (C) $\rightarrow$ (q), (D) $\rightarrow$ (r)
 (C) (A) $\rightarrow$ (q), (B) $\rightarrow$ (r), (C) $\rightarrow$ (p), (D) $\rightarrow$ (s)
 (D) (A) $\rightarrow$ (r), (B) $\rightarrow$ (p), (C) $\rightarrow$ (q), (D) $\rightarrow$ (s)

Q22 If S sum to 50 terms of $1 + (2)(3) + 4 + (5)(6) + 7 + (8)(9) + \dots$ then $S/9935$

Q23 Let $a_n = \frac{2^3-1}{2^3+1} \cdot \frac{3^3-1}{3^3+1} \cdot \frac{4^3-1}{4^3+1} \dots \frac{n^3-1}{n^3+1}$, then $\lim_{n \rightarrow \infty} 3a_n$ is

Q24 If the sides of a right angled triangle from an AP , the sines of the acute angles are

- (A) $\frac{3}{5}, \frac{4}{5}$
 (B) $\sqrt{3}, \frac{1}{3}$
 (C) $\sqrt{\frac{\sqrt{5}-1}{2}}, \sqrt{\frac{\sqrt{5}+1}{2}}$
 (D) $\frac{\sqrt{3}}{2}, \frac{1}{2}$

Q25 The sixth term of an AP is equal to 2. The value of the common difference of the AP which makes the product $a_1 a_4 a_5$ least, is
 (A) $8/5$ (B) $5/4$
 (C) $2/3$ (D) None of these

Q26 If the arithmetic progression whose common difference is non-zero, the sum of first $3n$ terms is equal to the sum of the next n terms. The ratio of the sum of the first $2n$ terms of the next $2n$ terms is
 (A) $1/5$ (B) $2/3$
 (C) $3/4$ (D) None of these

Q27 The coefficient of x^{n-2} in the polynomial $(x-1)(x-2)(x-3)\dots(x-n)$, is
 (A) $\frac{n(n^2+2)(3n+1)}{24}$
 (B) $\frac{n(n^2-2)(3n+2)}{24}$
 (C) $\frac{n(n^2+1)(3n+4)}{24}$
 (D) None of these

Q28 Consider the pattern shown below:

Row	1	1		
Row	2	3	5	
Row	3	7	9	11
Row	4	13	15	17, 19, etc

The number at the end of row 60 is

- (A) 3659 (B) 3519
 (C) 3681 (D) 3731

Q29 Let a_n be the n th term of an AP . If $\sum_{r=1}^{100} a_{2r} = \alpha$ and $\sum_{r=1}^{100} a_{2r-1} = \beta$, the common difference of the AP is
 (A) $\alpha - \beta$
 (B) $\beta - \alpha$
 (C) $\frac{\alpha - \beta}{2}$



(D) None of these

Q30 If a_1, a_2, a_3, a_4, a_5 are in HP, then $a_1 a_2 + a_2 a_3 + a_3 a_4 + a_4 a_5$ is equal to

- (A) $2a_1 a_5$
(B) $3a_1 a_5$
(C) $4a_1 a_5$
(D) $6a_1 a_5$

Q31 If a, b, c and d are four positive real numbers such that

$abcd = 1$, the minimum value of $(1+a)(1+b)(1+c)(1+d)$ is

- (A) 1 (B) 4
(C) 16 (D) 64

Q32 If a, b, c are in AP and $(a+2b-c)(2b+c-a)(c+a-b) = \lambda abc$, then λ is

- (A) 1 (B) 2
(C) 4 (D) None of these

Q33 If $a_1, a_2, a_3, \dots$ are in GP with first term a and common ratio r , then

$\frac{a_1 a_2}{a_1^2 - a_2^2} + \frac{a_2 a_3}{a_2^2 - a_3^2} + \frac{a_3 a_4}{a_3^2 - a_4^2} + \dots + \frac{a_{n-1} a_n}{a_{n-1}^2 - a_n^2}$ is equal to

- (A) $\frac{nr}{1-r^2}$
(B) $\frac{(n-1)r}{1-r^2}$
(C) $\frac{nr}{1-r}$
(D) $\frac{(n-1)r}{1-r}$

Q34 The sum of the first ten terms of an AP is four times the sum of the first five terms, the ratio of the first term to the common difference is

- (A) $1/2$ (B) 2
(C) $1/4$ (D) 4

Q35 If $\cos(x-y), \cos x$ and $\cos(x+y)$ are in HP, the

$\cos x \sec\left(\frac{y}{2}\right)$ is equal to

- (A) $\pm\sqrt{2}$
(B) $\frac{1}{\sqrt{2}}$
(C)

$$-\frac{1}{\sqrt{2}}$$

(D) None of these

Q36 If 11 AMs are inserted between 28 and 10, the number of integral AMs is

- (A) 5 (B) 6
(C) 7 (D) 8

Q37 If x, y, z are in GP ($x, y, z > 1$), then $\frac{1}{2x+\ln x}, \frac{1}{4x+\ln y}, \frac{1}{6x+\ln z}$ are in W.

- (A) AP
(B) GP
(C) HP
(D) None of these

Q38 The minimum value of the quantity $\frac{(a^2+3a+1)(b^2+3b+1)(c^2+3c+1)}{abc}$ where $a, b, c \in \mathbb{R}^+$, is

- (A) $\frac{11^3}{2^3}$
(B) 125
(C) 25
(D) 27

Q39 Suppose, α, β are roots of $ax^2 + bx + c = 0$ and γ, δ are roots of $Ax^2 + Bx + C = 0$.

If $\alpha, \beta, \gamma, \delta$ are in AP, then common difference of AP is

- (A) $\frac{1}{4}\left(\frac{b}{a} - \frac{B}{A}\right)$
(B) $\frac{1}{3}\left(\frac{b}{a} - \frac{B}{A}\right)$
(C) $\frac{1}{2}\left(\frac{b}{a} - \frac{B}{A}\right)$
(D) $\frac{1}{3}\left(\frac{c}{a} - \frac{C}{A}\right)$

Q40 Suppose, α, β are roots of $ax^2 + bx + c = 0$ and γ, δ are roots of $Ax^2 + Bx + C = 0$.

If a, b, c are in GP as well as $\alpha, \beta, \gamma, \delta$ are in GP, then A, B, C are in

- (A) AP only
(B) GP only
(C) AP and GP
(D) None of these

Q41



Column-I		Column-II	
(A)	If $4a^2 + 9b^2 + 16c^2 = 2(3ab + 6bc + 4ca)$, where a, b, c are non-zero numbers, then a, b, c are in	(P)	AP
(B)	If $(a^2 + b^2 + c^2)p^2 - 2p(ab + bc + cn) + (n^2 + b^2 + c^2) \leq 0$, where a, b, c, p, n are non-zero numbers, then a, b, c are	(Q)	GP
(C)	If $a^2 + 9b^2 + 25c^2 = abc\left(\frac{15}{a} + \frac{5}{b} + \frac{3}{c}\right)$, where a, b, c are non-zero numbers, then a, b, c are in	(R)	HP

- (A) $(A) \rightarrow (P), (B) \rightarrow (P), (C) \rightarrow (R)$
 (B) $(A) \rightarrow (R), (B) \rightarrow (Q), (C) \rightarrow (R)$
 (C) $(A) \rightarrow (R), (B) \rightarrow (R), (C) \rightarrow (Q)$
 (D) $(A) \rightarrow (R), (B) \rightarrow (R), (C) \rightarrow (P)$

Q42 Match the following

Column-I		Column-II	
(A)	For an AP $a_1, a_2, a_3, \dots, a_n, \dots, a_1 = \frac{5}{2}; a_{10} = 16$. If $a_1 + a_2 + \dots + a_n = 110$, then 'n' equals	(P)	9
(B)	The interior angles of a convex non-equilateral polygon of 9 sides are in AP. The least positive integer that limits the upper value of the common difference between the measures of the angles in degrees is	(Q)	10
(C)	For an increasing GP, $a_1, a_2, a_3, \dots, a_n, \dots, a_6 = 4a_4; a_9 - a_7 = 192$, if $a_4 + a_5 + \dots + a_n = 1016$, then n equals	(R)	11
		(S)	12

- (A) $(A) \rightarrow (Q), (B) \rightarrow (P), (C) \rightarrow (R)$
 (B) $(A) \rightarrow (P), (B) \rightarrow (Q), (C) \rightarrow (R)$
 (C) $(A) \rightarrow (R), (B) \rightarrow (P), (C) \rightarrow (Q)$
 (D) $(A) \rightarrow (R), (B) \rightarrow (R), (C) \rightarrow (P)$

Q43 Let a, b, c, d be positive real numbers with $a < b < c < d$. Given that a, b, c, d are the first four terms of an AP and a, b, d are in GP. The value of $\frac{ad}{bc}$ is $\frac{p}{q}$, where p and q are prime numbers, then the value of q is

Q44 If the coefficient of x in the expansion of $\prod_{r=1}^{110} (1 + rx)$ is $\lambda(1 + 10)(1 + 10 + 10^2)$, then the value of λ is

Q45 If $(1^2 - t_1) + (2^2 - t_2) + \dots + (n^2 - t_n) = \frac{n(n^2 - 1)}{3}$

then t_n is equal to

- (A) n^2
 (B) $2n$
 (C) $n^2 - 2n$
 (D) none of these

Q46 Let $S_n = \sum_{r=1}^n t_r = \frac{1}{6}n(2n^2 + 9n + 13)$, then $\sum_{r=1}^n \sqrt{t_r}$ equals

- (A) $\frac{1}{2}n(n+1)$
 (B) $\frac{1}{2}n(n+3)$
 (C) $(n+1)^2$
 (D) n^2

Q47 Let $f: R \rightarrow R$ be a function such that $f(1) = 2$ and it satisfies the relation $f(x+y) = f(x)f(y)$ for all natural numbers x and y . Then the value of the natural number a such that $\sum_{K=1}^n f(a+K) = 16(2^n - 1)$ is

(A) 3 (B) 4
 (C) 5 (D) 6

Q48 The value of $\sum_{n=3}^{\infty} \frac{1}{n^5 - 5n^3 + 4n} =$

(A) $\frac{1}{120}$
 (B) $\frac{1}{96}$
 (C) $\frac{1}{24}$
 (D) $\frac{1}{144}$

Q49 If n is a positive integer satisfying the equation

$$2 + (6 \cdot 2^2 - 4 \cdot 2) + (6 \cdot 3^2 - 4 \cdot 3) + \dots + (6 \cdot n^2 - 4 \cdot n) = 140$$

then the value of n is

Q50 The sequence $a_1, a_2, a_3, \dots$ is a geometric sequence with common ratio r . The sequence $b_1, b_2, b_3, \dots$ is also a geometric sequence. If $b_1 = 1, b_2 = \sqrt[4]{7} - \sqrt[4]{28} + 1, a_1 = \sqrt[4]{28}$ and $\sum_{n=1}^{\infty} \frac{1}{a_n} = \sum_{n=1}^{\infty} b_n$, then the value of $(1 + r^2 + r^4)$ is



Answer Key

Q1 (D)
Q2 (A)
Q3 (B)
Q4 (A)
Q5 (C)
Q6 (D)
Q7 (A)
Q8 (D)
Q9 (A)
Q10 (C)
Q11 (B)
Q12 (C)
Q13 (A)
Q14 (D)
Q15 (A)
Q16 (A)
Q17 (A)
Q18 (D)
Q19 (C)
Q20 (C)
Q21 (B)
Q22 5
Q23 2
Q24 (A)
Q25 (C)

Q26 (A)
Q27 (D)
Q28 (A)
Q29 (D)
Q30 (C)
Q31 (C)
Q32 (C)
Q33 (B)
Q34 (A)
Q35 (A)
Q36 (A)
Q37 (C)
Q38 (B)
Q39 (A)
Q40 (B)
Q41 (B)
Q42 (C)
Q43 3
Q44 5
Q45 (D)
Q46 (B)
Q47 (A)
Q48 (B)
Q49 4
Q50 7



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